

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION
प्री-बोर्ड परीक्षा / PRE-BOARD EXAMINATION – 2024-25

कक्षा / CLASS –X

अधिकतम अंक/MAX. MARKS –50

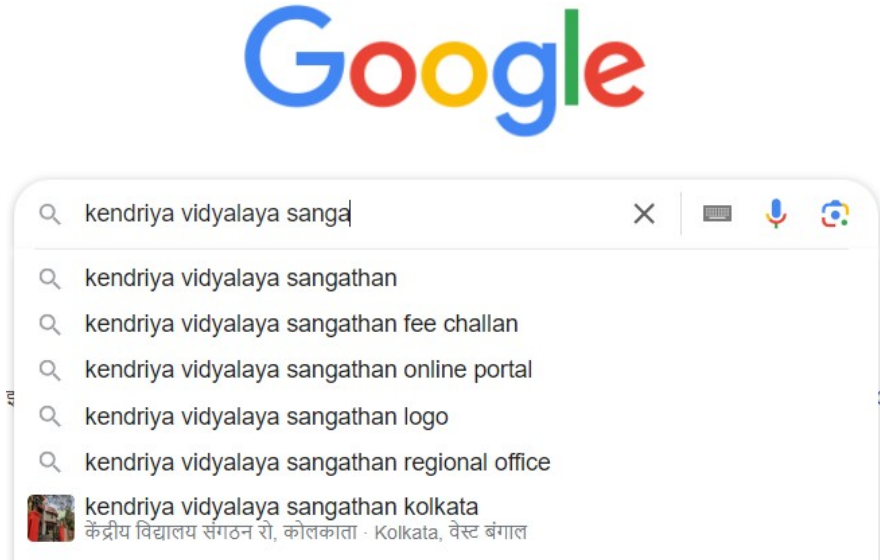
विषय /SUB. -Artificial Intelligence

समय/TIME – 02 घंटे/Hours

MARKING SCHEME

	Note-Correct options in MCQ are marked underlined and Bold	
	<u>SECTION A: OBJECTIVE TYPE QUESTIONS</u>	
Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	Pranjali gets up at 5 am and goes to her badminton classes. Then she comes home and finishes her homework before going to school. She does this all by herself. No one tells her to do it. This is an example of <u>a) Self-motivation</u> b) External motivation c) Both self and external motivation d) Not any specific type of motivation	1
ii.	_____ act as an interface between the user and the computer. <u>a) Operating System</u> b) MS Excel c) Impress d) Digital Documentation	1
iii.	Which of the following are misconception about Entrepreneur? a) Entrepreneurs are born, not made. b) A person having a big business is an entrepreneur. c) A person needs a lot of money to start a business. <u>d) All of the above</u>	1
iv.	SMART method can be used to set goals to make you successful in your career and personal life. What does 'M' in SMART stand for? a) Motivator <u>b)Measurable</u> c) Maintainable d)Mentor	1
v.	Which of these is NOT a common communication barrier? a) Linguistic barrier b) Interpersonal barrier <u>c) Financial barrier</u> d) Organizational barrier	1
vi.	Choose the option which defines sustainable development. a) Taking care of future generations. b) Taking care of only ourselves. <u>c) Taking care of ourselves and the future generations.</u> d) None of the above	1
Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Which of the following statement is incorrect?	1

	a. Email Filtering is an application of NLP b. Price comparison websites are applications of Data Science c. Virtual Assistants are not applications of NLP d. Self-driving cars are applications of Computer Vision.	
ii.	Results based on discriminatory assumptions made while developing algorithm by the developer is called as _____. a) AI Ethics b) AI Bias c) Test Data d) Training Data	1
iii.	The dataset is divided in two parts _____. a) Machine dataset & Model dataset b) Training dataset & Test dataset c) Global dataset & local dataset d) None of the above	1
iv.	Which of the following refers to “Where” among 4Ws canvas? a. stakeholders b. nature of the problem c. context/situation/location d. solution or benefits to the stakeholders	1
v.	Which industry uses data science to analyze customer profiling and past expenditures? a) Genetics & Genomics b) Fraud and Risk Detection c) Internet Search d) Airline Route Planning	1
vi.	Aaron is facing issue in his internet connectivity at home. He called customer care service, A virtual assistant took his complaint and replied with an acknowledgement number. What is the AI technology used by Internet service provider? a) Data Science b) Machine Learning c) Computer Vision d) Natural Language Processing	1
Q. 3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	_____ is the ability to recognize and create sounds, rhythm and sound pattern. a) Mathematical Logical Reasoning b) Musical Intelligence c) Spatial Visual Intelligence d) Linguistic Intelligence	1
ii.	The method of collecting correct and dependable data to work with is known as _____. a) Problem Scoping b) Data Acquisition c) Data Exploration d) Modeling	1

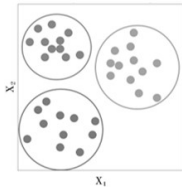
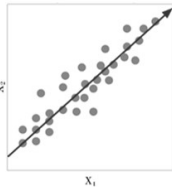
iii.	What is the purpose of website recommendations? a) To improve user experience b) To promote products c) To suggest similar products <u>d) All of the above</u>	1
iv.	What is the primary goal of Computer Vision? a) To enhance image quality <u>b) To interpret and make decisions based on visual data</u> c) To store images efficiently d) To increase resolution of images	1
v.	What is a chatbot called which uses simple FAQs without any intelligence? a) Smart Chatbot <u>b) Script Chatbot</u> c) AI Chatbot d) ML Chatbot	1
vi.	The process of understanding the reliability of any AI model based on output by feeding the test dataset is a) Data feed b) Data Reliability <u>c) Model Evaluation</u> d) None of these	1
Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Based on the image below, choose the correct domain or domains of AI required for the google search engine:  a) Data Science b) NLP c) Computer Vision <u>d) All of the above</u>	1
ii.	The percentage of true positive cases versus all the cases where the prediction is true is defined as	1

	a) Precision b) Accuracy c) F1 Score d) None of these	
iii.	What does CSV stand for? a) Comma separated values b) Computer spreadsheet values c) Centralized storage volumes d) Compressed structured variables	1
iv.	What does segmentation in Computer Vision refer to? a) Enhancing image details b) Dividing an image into parts or regions c) Reducing image size d) Increasing image resolution	1
v.	_____are the words which occur very frequently in the corpus but do not add any value to it. a) Special Characters b) Stopwords c) Roman Numbers d) Useless Words	1
vi.	The measure of balance between precision and recall. a. Accuracy b. F1 Score c. Precision d. None of these	1
Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	What is important to make a Good AI Machine? a) Algorithm b) Data c) Test cases d) All the above.	1
ii.	The F1 score ranges from _____to _____ a) -1,0 b) 0,1 c) -1,1 d) None of the above	1
iii.	Assertion (A): Computer vision is a field of artificial intelligence that enables computers to interpret and make decisions based on visual data from the world. Reasoning (R): Computer vision uses algorithms to process and analyse images and videos, enabling tasks like object detection and facial recognition. a) Both A and R are true, and R is the correct explanation for A. b) Both A and R are true, but R is not the correct explanation for A. c) A is true, but R is false. d) A is false, but R is true.	1
iv.	What do we call the process of dividing a string into component words?	1

	a) Regression <u>b) Tokenization</u> c) Classification d) Clustering	
v.	TF-IDF in NLP is defined as: a) Term Frequency and Definite Frequency b) Term Frequency and Indefinite Frequency <u>c) Term Frequency and Inverse Document Frequency</u> d) Term Frequency and Integrated Document Frequency	1
vi.	Statement 1: F1 score is evaluated based on precision or recall. Statement 2: When the F1 score is 0, the model accuracy is 100% a) Both statement1 and statement 2 are correct. b) Both statement1 and statement 2 are incorrect. <u>c) Statement 1 is correct but statement 2 is incorrect.</u> d) Statement 1 is incorrect but statement 2 is correct.	1
<u>SECTION B: SUBJECTIVE TYPE QUESTIONS</u>		
Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.		
Q. 6	Mention any two positive facial expressions which you can use in making effective communication. Ans) • Smiling when meeting someone. • Keeping face relaxed. • Matching expressions with your words. • Nodding while listening. (Any two Points)	2
Q. 7	What is the importance of setting goals in life? Ans) Following are the importance of Goal setting1. Goals allow you to separate out what's important. 2. It helps you to focus on the end result instead of less Important work. 3. This will make you successful in your Career and personal life.	2
Q. 8	Ravi wants to change his phone but he wants to transfer his old data for later use so suggest him the term regarding this process. Ans) The process through which Ravi can transfer his old data for later use is known as data backup. Backing up data means to save the information present on your computer on another device, such as CD/DVD drives or hard disk.	2
Q. 9	List two qualities of successful Entrepreneurs. Ans) They are confident. They believe in themselves and their abilities. • They keep trying new ideas in their business. • They are patient. • They are creative and think differently about business ideas. • They take responsibility for their actions. (Any two Points)	2
Q. 10	How can we become responsible about our own environment?	2

	Ans) We can become responsible about our own environment by 1. reusing paper, glass, plastic, water, etc. 2. taking cloth bags to market carrying fruits and vegetables. 3. donate things we do not use such as clothes, books, furniture, food, etc. 4. Buy and eat seasonal fruits and vegetables from local growers. 5. Repair leaking taps and pipes to avoid wasting water. 6. Sort and treat garbage before disposing.	
Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)		
Q. 11	What is AI Bias? Ans) AI Bias is an irregularity in the outcome of a model/Algorithm due to data collected is unbalanced and based on wrong assumptions.	2
Q. 12	Rajesh wants to group students data based on the grades obtained in the examination. Which AI model should he use and why? Ans) Rajesh should use Classification model because Classification models are a type of machine learning model that divides data points into predefined groups i.e. grades in examination.	2
Q. 13	Rohan wants to access data from different sources for his new AI project. Suggest him any two ways to collect the data for his new project. Also explain Rohan about the role of data in AI project. 1. Data which is available for public usage only should be taken up. 2. Personal datasets should only be used with the consent of the owner. 3. One should never breach someone's privacy to collect data. 4. Data should only be taken from reliable sources as the data collected from random sources can be wrong or unusable. 5. Reliable sources of data ensure the authenticity of data which helps in the proper training of the AI model. 6. Data should be relevant to the problem. (1 mark for Any 02 points) (1 marks for describing the role of data) Data is essential to artificial intelligence (AI) projects because it provides the information that AI systems use to learn, make decisions, and improve over time. The quality of the data used to train AI models directly impacts their performance, reliability, and fairness.	2
Q. 14	Explain the difference between grayscale and RGB images. Ans) Grayscale images consist of shades of gray, ranging from black to white, with each pixel representing an intensity value. RGB images use three color channels (Red, Green, Blue), where each pixel is a combination of these three colors, allowing for a wide range of colors in the image	2
Q. 15	Define the following: a) Stemming b) Lemmatization Ans) Stemming: Stemming is a rudimentary rule-based process of stripping the suffixes ("ing", "ly", "es", "s" etc) from a word. Stemming is a process of reducing words to their word stem, base or root	2

	form (for example, books — book, looked — look). Lemmatization: Lemmatization is an organized & step by step procedure of obtaining the root form of the word, it makes use of vocabulary (dictionary importance of words) and morphological analysis (word structure and grammar relations). (1 mark for each definition)														
Q. 16	Draw the confusion matrix for the following data. the number of true positive = 80 the number of true negative 17 the number of false positive = 72 the number of false negative = 110 Ans) <table border="1"><tr><th colspan="2" rowspan="2">Confusion Matrix:</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr><tr><th rowspan="2">Prediction</th><th>Yes</th><td>80</td><td>72</td></tr><tr><th>No</th><td>110</td><td>17</td></tr></table>	Confusion Matrix:		Reality		Yes	No	Prediction	Yes	80	72	No	110	17	2
Confusion Matrix:				Reality											
		Yes	No												
Prediction	Yes	80	72												
	No	110	17												
Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)															
Q. 17	Rahul wants to pursue the AI course after class XII. His parents didn't know much about its domains. Explain them the domains of AI. 1 mark for Data Sciences, CV & NLP 3 marks for explanation of each domain	4													
Q. 18	Explain the 4Ws problem canvas and explain each one of them briefly. Ans)The 4Ws problem canvas is the basic template while scoping a problem and using this canvas, the picture becomes clearer while we are working to solve it. a) Who: The “Who” block helps you in analyzing the people getting affected directly or indirectly due to it? Under this, you find out who the ‘stakeholders’ to this problem are and what you know about them. Stakeholders are the people who face this problem and would be benefitted with the solution. b) What: Under the “What” block, you need to look into what you have on hand. At this stage, you need to determine the nature of the problem. What is the problem and how do you know that it is a problem? c) Where: In this block, you need to focus on the context/situation/location of the problem. It will help you look into the situation in which the problem arises, the context of it, and the locations where it is prominent. d) Why: in the “Why” canvas, think about the benefits which the stakeholders would get from the solution and how would it benefit them as well as the society (1 mark for each ‘W’)	4													
Q. 19	Identify and explain the two types of the AI model in the figures given	4													

	below.																																																					
	Figure 1  Figure 2  Ans) 1 mark for identification and 3 marks for explanation Figure 1-Clustering Model- Clustering is an unsupervised machine learning technique designed to group unlabeled examples based on their similarity to each other Figure 2-Regression Model -Regression is a type of supervised learning which is used to predict continuous value. Example: Regression is used to predict the weather. it is also used widely for weather forecasting.																																																					
Q. 20	Create a document vector table for the following documents Document 1: Aman and Anil are stressed Document 2: Aman went to a therapist Document 3: Anil went to download a health chatbot Ans) <table><tr><th></th><th>aman</th><th>and</th><th>anil</th><th>are</th><th>stressed</th><th>went</th><th>to</th><th>a</th><th>therapist</th><th>download</th><th>health</th><th>chatbot</th></tr><tr><td>Document 1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Document 2</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Document 3</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>		aman	and	anil	are	stressed	went	to	a	therapist	download	health	chatbot	Document 1	1	1	1	1	1	0	0	0	0	0	0	0	Document 2	1	0	0	0	0	1	1	1	1	0	0	0	Document 3	0	0	1	0	0	1	1	1	0	1	1	1	4
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Q. 21	The country was shaken up by a series of flood which has done a huge damage to the people as well as the infrastructure. To address this issue, an AI model has been created which can predict if there is a chance of a flood or not. The confusion matrix for the same is as below: calculate Accuracy, precision, recall and F1 score. <table><tr><th colspan="2" rowspan="2">Confusion Matrix:</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr><tr><th rowspan="2">Prediction</th><th>Yes</th><td>90</td><td>10</td></tr><tr><th>No</th><td>10</td><td>50</td></tr></table> Accuracy=(90+50)/160=0.875 Recall=90/100=0.9 Precision=90/100=0.9 F1 score=2*recall*precision/(recall+precision) =2*0.81/1.8=0.9 (1 mark for each)	Confusion Matrix:		Reality		Yes	No	Prediction	Yes	90	10	No	10	50	4																																							
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